

4(a)(i)	1 crime scene DNA may be present in, small amount / minute quantity ; 2 PCR, amplifies / replicates / copies / multiplies, DNA ;	2
4(a)(ii)	1 (negative) DNA moves to, positive electrode / anode ; 2 smaller / lighter, fragments move, faster / further / longer distance ;	2

Question	Answer	Marks
4(b)(i)	any two from: 1 (GEDmatch) database is large(r than police database) ; 2 police database only covers people, convicted of / suspected of, crime ; 3 GEDmatch has data for, innocent people / people with no reason to hide their DNA ; 4 provides multiple leads to guilty people who are hiding their DNA ;	2
4(b)(ii)	any two from: 1 store of / large quantity of / contains, biological / DNA / genome, data / information ; 2 for, analysis / processing ; 3 by, computer / software / algorithm / statistics ;	2
4(b)(iii)	any two from: 1 help police / catch criminals / imprison criminals ; 2 make, society / community, safe(r) ; 3 <i>ref. to</i> DNA breakthroughs help, justice / law and order ; 4 sharing own data also shares relations' data without their consent ; 5 AVP ;	2